Physics 4C

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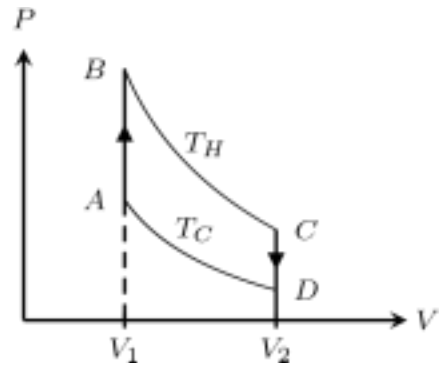
1. The heat cycle represented in the following *PV* diagram is known as the Stirling cycle. It consists of two isochoric processes (a-b and c-d) and two isothermal processes (b-c and d-a).




An ideal gas with $\gamma = 1.4$ is put through the cycle. If $V2/V1 = 5.0$ and $T_L/TH = 0.75$, what is the efficiency?

- $e = \frac{Q_H - Q_L}{Q_H} = \frac{W}{Q_H}$

from $B \to C$, $PV = NkT_H \Rightarrow P = \frac{NkT_H}{V}$

from $D \to A$, $PV = NkT_L \Rightarrow P = \frac{NkT_L}{V}$

- $W_{B\to C} + W_{D\to A} = \int_{V_1}^{V_2} \frac{NkT_H}{V} dV + \int_{V_2}^{V_1} \frac{NkT_L}{V} dV$

$= NkT_H \int_{V_1}^{V_2} \frac{1}{V} dV - NkT_L \int_{V_1}^{V_2} \frac{1}{V} dV$

$= NkT_H \ln\left|\frac{V_2}{V_1}\right| - NkT_L \ln\left|\frac{V_2}{V_1}\right|$

$= Nk \ln\left|\frac{V_2}{V_1}\right| (T_H - T_L)$

- $Q_H = Q_{A\to C} = Q_{A\to B} + Q_{B\to C}$
- $Q_{A\to B} = N c_V \Delta T = N c_V (T_H - T_L)$
- $Q_{B\to C} = W_{B\to C} = NkT_H \ln\left|\frac{V_2}{V_1}\right|$
- $Q_{A\to C} = N c_V (T_H - T_L) + NkT_H \ln\left|\frac{V_2}{V_1}\right|$
- $e = \frac{W}{Q_H} = \frac{Nk \ln\left|\frac{V_2}{V_1}\right| (T_H - T_L)}{N c_V (T_H - T_L) + NkT_H \ln\left|\frac{V_2}{V_1}\right|}$

$= \frac{Nk \ln 5 (T_H - 0.75T_H)}{\frac{5}{2}Nk(T_H - 0.75T_H) + NkT_H \ln 5}$

$= \frac{\frac{1}{4}Nk \ln 5\, T_H}{\frac{1}{4}\cdot\frac{5}{2}NkT_H + NkT_H \ln 5} = \frac{\frac{1}{4}\ln 5}{\frac{5}{8} + \ln 5} = 0.18007 = \boxed{18.007\%}$

2. A Stirling engine operates between a hot reservoir at temperature *TH* = 400 K and a cold reservoir at temperature *TC* = 300 K. The working substance is *n* = 0.15 mol of ideal gas with gamma factor $\gamma = 1.4$. Assume all heat going into the engine comes from the hot reservoir and all heat dissipated by the engine goes to the cold reservoir and compute the change of entropy for each cycle.

Using variables set up by previous problem,

$\Delta S_H = \frac{Q_H}{T_H} = \frac{N c_V (T_H - T_L) + \cancel{NkT_H \ln\left|\frac{V_2}{V_1}\right|}}{T_H} = \frac{\frac{5}{2}Nk(T_H - T_L)}{T_H} = \frac{\frac{5}{2}nR(T_H - T_L)}{T_H}$

0, since $\Delta V = 0$, so $\frac{V_2}{V_1} = 1$.

$\Delta S_L = \frac{Q_L}{T_L} = \frac{\frac{5}{2}nR(T_H - T_L)}{T_L}$

$\Delta S = \Delta S_H - \Delta S_L = \frac{\frac{5}{2}nR(T_H - T_L)}{T_H} - \frac{\frac{5}{2}nR(T_H - T_L)}{T_L} = \frac{5}{2}nR(T_H - T_L)\left(\frac{1}{T_H} - \frac{1}{T_L}\right)$

$= \frac{5}{2}(0.15\,\text{mol})\left(8.314\,\frac{J}{mol\cdot K}\right)(100\,K)\left(\frac{1}{400\,K} - \frac{1}{300\,K}\right)$

$= \boxed{0.26\,\frac{J}{K}}$

3.

At point $A$ in a Carnot cycle, 2.34 mol of a monatomic ideal gas has a pressure of 1 400 kPa, a volume of 10.0 L, and a temperature of 720 K. The gas expands isothermally to point $B$ and then expands adiabatically to point $C$, where its volume is 24.0 L. An isothermal compression brings it to point $D$, where its volume is 15.0 L. An adiabatic process returns the gas to point $A$. (a) Determine all the unknown pressures, volumes, and temperatures as you fill in the following table:

| | $P$ | $V$ | $T$ |
|---|---|---|---|
| $A$ | 1 400 kPa | 10.0 L | 720 K |
| $B$ | 875 kPa | 16.0 L | 720 K |
| $C$ | 445 kPa | 24.0 L | 549 K |
| $D$ | 712 kPa | 15.0 L | 549 K |

(b) Find the energy added by heat, the work done by the engine, and the change in internal energy for each of the steps $A \to B$, $B \to C$, $C \to D$, and $D \to A$. (c) Calculate the efficiency $W_{net}/|Q_h|$. (d) Show that the efficiency is equal to $1 - T_C/T_A$, the Carnot efficiency.

· between points A and D,

$P_A V_A^{5/3} = P_D V_D^{5/3}$

$P_D = \frac{P_A V_A^{5/3}}{V_D^{5/3}} = P_A\left(\frac{V_A}{V_D}\right)^{5/3} = (1400\,kPa)\left(\frac{10\,L}{15\,L}\right)^{5/3}$

$= 712\,kPa$

b) · from $A \to B$, $\Delta E_{int} = 0$, $Q = W = nRT_H \ln\left(\frac{V_B}{V_A}\right) = \boxed{6580\,J}$

· from $B \to C$, $\Delta Q = 0$, $\Delta E_{int} = W = \frac{3}{2}nR(T_L - T_H) = \boxed{-4990\,J}$

· from $C \to D$, $\Delta E_{int} = 0$, $Q = W = nRT_L \ln\left(\frac{V_B}{V_A}\right) = \boxed{-5020\,J}$

· from $D \to A$, $\Delta Q = 0$, $\Delta E_{int} = W = \frac{3}{2}nR(T_H - T_L) = \boxed{4990\,J}$

c) $e = \frac{W_{net}}{|Q_H|} = \frac{(6580 - 4990 - 5020 + 4990)\,J}{6580\,J} = \boxed{23.7\%}$

d) $1 - \frac{T_L}{T_A} = 1 - \frac{549}{720} = 0.2375 = \boxed{23.75\%}$